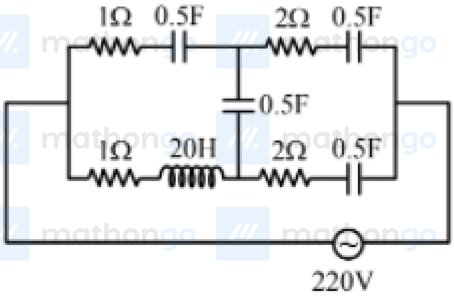
Q1 2021 (31 Aug Shift 2)

At very high frequencies, the effective impendance of the given circuit will be .... Ω



Q2 2021 (31 Aug Shift 1)

In an ac circuit, an inductor, a capacitor and a resistor are connected in series with $X_L = R = X_C$ Impedance of this circuit is:

(1) $2R^2$

(2) Zero

(3) R

(4) $R\sqrt{2}$

Q3 2021 (27 Aug Shift 2)

An ac circuit has an inductor and a resistor of resistance R in series, such that $X_L = 3R$. Now, a capacitor is added in series such that $X_C = 2R$. The ratio of new power factor with the old power factor of the circuit is $\sqrt{5} : x$. The value of x is

Q4 2021 (27 Aug Shift 1)

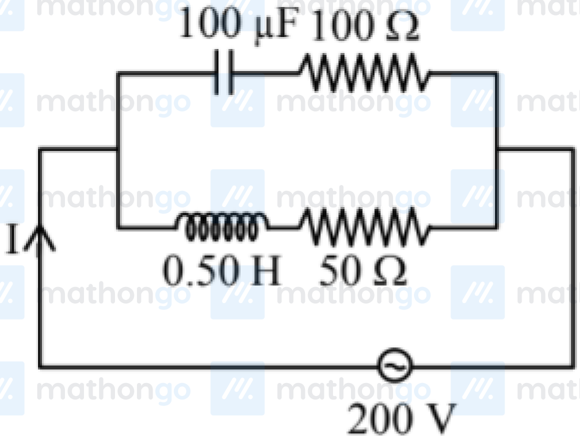
The alternating current is given by

$$i = \left\{\sqrt{42}\sin\left(\frac{2\pi}{T}t\right) + 10\right\} A$$

The r.m.s. value of this current is ...... A.

Q5 2021 (26 Aug Shift 2)

In the given circuit the AC source has $\omega = 100 \text{rads}^{-1}$. Considering the inductor and capacitor to be ideal, what will be the current I flowing through the circuit?



(1) 5.9 A

(2) 4.24 A

(3) 0.94 A

(4) 6 A

Q6 2021 (26 Aug Shift 1)

A series LCR circuit driven by 300 V at a frequency of 50 Hz contains a resistance $\mathrm{R} = 3 \mathrm{k}\Omega$ an inductor of inductive reactance $X_L = 250\pi\Omega$ and an unknown capacitor. The value of capacitance to maximize the average power should be : (Take $\pi^2 = 10$)

(1) 4μF

(2) 25μF

(3) 400μF

(4) 40μF

Q7 2021 (26 Aug Shift 1)

An inductor coil stores $64 \mathrm{~J}$ of magnetic field energy and dissipates energy at the rate of $640 \mathrm{~W}$ when a current of $8 \mathrm{~A}$ is passed through it. If this coil is joined across an ideal battery, find the time constant of the circuit in seconds :

(1) 0.4

(2) 0.8

(3) 0.125

(4) 0.2

Answer Key

Q1 (2)	**Q2** (3)	**Q3** (1)	**Q4** (11)
Q5 (2)	**Q6** (1)	**Q7** (4)	

Q1 (2)

$$X_L = 2\pi fI$$

f is very large

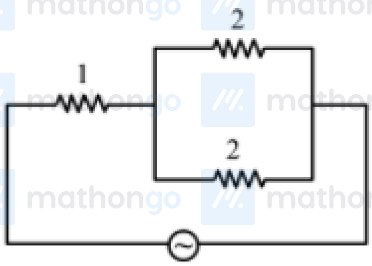
$\therefore X_L$ is very large hence open circuit.

$$X_C = \frac{1}{2\pi fC}$$

f is very large.

$\therefore X_C$ is very small, hence short circuit.

Final circuit



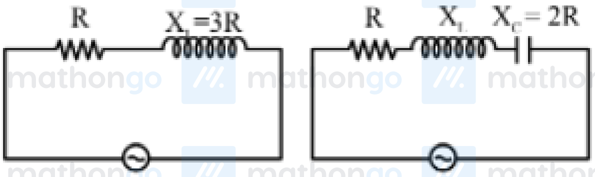
$Z_{eq} = 1 + \frac{2\times2}{2+2} = 2$

Q2 (3)

$$Z = \sqrt{(X_L - X_C)^2 + R^2} = R \quad \because X_L = X_C$$

Option (3)

Q3 (1)



$\cos\phi=\frac{\mathrm{R}}{\sqrt{\mathrm{R}^2+3\mathrm{R}^2}}\quad \cos\phi'=\frac{\mathrm{R}}{\sqrt{\mathrm{R}^2+\mathrm{R}^2}}$

$=\frac{1}{\sqrt{10}}\quad =\frac{1}{\sqrt{2}}$

$\frac{\cos\phi'}{\cos\phi}=\frac{\sqrt{10}}{\sqrt{2}}=\frac{\sqrt{5}}{1}\quad \therefore \mathrm{x}=1$

Q4 (11)

$\mathrm{f}_{\mathrm{rms}}^2=\mathrm{f}_{1\,\mathrm{ms}}^2+\mathrm{f}_{2\,\mathrm{ms}}^2$

$=\left(\frac{\sqrt{42}}{\sqrt{2}}\right)^2+10^2$

$=121\Rightarrow \mathrm{f}_{\mathrm{ms}}=11\ \mathrm{A}$

Q5 (2)

$Z_C=\sqrt{\left(\frac{1}{\omega C}\right)^2+R^2}$

$=\sqrt{\left(\frac{1}{100\times100\times10^{-6}}\right)^2+100^2}$

$Z_c=\sqrt{(100)^2+(100)^2}$

$=100\sqrt{2}$

$Z_L=\sqrt{(\omega L)^2+R^2}$

$\sqrt{(100\times0.5)^2+50^2}$

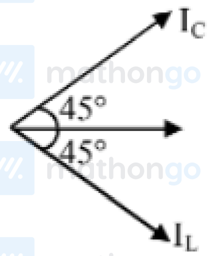
$=50\sqrt{2}$

$i_c=\frac{200}{z_C}=\frac{200}{100\sqrt{2}}=\sqrt{2}$

$i_L=\frac{200}{z_{I\cdot}}=\frac{200}{50\sqrt{2}}=2\sqrt{2}$

$\cos\phi_1=\frac{100}{10\sqrt{2}}=\frac{1}{\sqrt{2}}\Rightarrow\phi_1=45^\circ$

$\cos\phi_2=\frac{50}{50\sqrt{2}}=\frac{1}{\sqrt{2}}\Rightarrow\phi_2=45^\circ$



$$I=\sqrt{I_C^2+I_L^2}$$
$$=\sqrt{2+8}$$
$$=\sqrt{10}$$
$$\mathrm{I}=3.16\ \mathrm{A}$$

Q6 (1)

For maximum average power

$$X_L=X_C$$

$250\pi=\frac{1}{2\pi(50)C}$

$$C=4\times10^{-6}$$

Option (1)

Q7 (4)

$U=\frac{1}{2}Li^2=64\Rightarrow L=2$
$i^2R=640R=\frac{640}{(8)^2}=10$

$$\tau=\frac{L}{R}=\frac{1}{5}=0.2$$

Option (4)